

Q.1 What is geometric intuition behind SVM?

→ The main intention is to find ~~a~~ the optimal boundary - hyperplane, that is the best line which separates data points of classes while maximizing margin between them.

Q.2 How would you explain convex hull in light of SVM

→ We simply build a convex hull for class A and B and draw a perpendicular on the shortest distance between closest points of both these hulls.

Q.3 What do we know about hard margin SVM and soft margin SVM?

→ If a point a satisfies the equation $(w \cdot a + b) \geq 1$ then a is correctly classified. So if the points are linearly separable then only the hyperplane is able to distinguish between them.

Q.4 What is hinge loss?

→ Hinge loss is a loss function which penalizes the SVM model for inaccurate predictions.

If $y_i(w \cdot a + b) \geq 1$, hinge loss is '0' i.e. points are correctly classified, when $y_i(w \cdot a + b) < 1$, hinge loss increases massively.

As $y_i(w \cdot a + b)$ increases with every misclassified point, the upper bound of hinge loss $\{1 - y_i(w \cdot a + b)\}$ also increases exponentially.